

Program: Diploma in Electrical Engineering	
Course Code: 4541	Course Title: Power Electronics
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To introduce students to the basic theory of power semiconductor devices and passive components, and their practical applications in power electronics.
- To familiarize students with the principle of operation of different power electronic converters and their applications.
- To provide a strong foundation for further study of power electronics

Course Prerequisites:

Topic/Description	Course code	Course Title	Semester
Basics of electronics	2031	Fundamentals of Electrical & Electronics Engineering	2
Electronic circuits	3031	Analog & Digital Circuits	3

Course Outcomes

On completion of the course, the students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Select an appropriate power semiconductor device for a particular application.	16	Understanding
CO2	Summarize the principle and operation of controlled rectifiers.	14	Understanding
CO3	Summarize the principle and operation of DC and AC converters.	12	Understanding
CO4	Summarize various inverters and electric drives.	16	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

On completion of the course, the students will be able to:

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Select appropriate power semiconductor device for a particular application.		
M1.01	Summarize the operation of power electronic devices.	8	Understanding
M1.02	Explain the operation of SCR	3	Understanding
M1.03	Classify and explain turn on and turn off methods of SCR.	4	Understanding
M1.04	Summarize the idea of datasheets of SCR	1	Understanding

Contents:

Power Electronic Devices- List various devices -Symbols (MOSFET, IGBT,GTO, LASCR, SCS, UJT, DIAC, TRIAC and SCR) -MOSFET - list different types -structure and working of N-channel enhancement type-IGBT--structure-working principle - applications-UJT- schematic diagram and working-DIAC & TRIAC-schematic diagram- working- V/I characteristics.

Silicon Controlled Rectifier (SCR)-structure-V-I characteristics- definitions-holding current- latching current

SCR Turn on and Turn off Methods -SCR Turn-On methods -list various methods - basic concepts - gate triggering -forward voltage triggering-resistance triggering- temperature triggering-light triggering- dv/dt triggering-SCR Turn-Off methods-list various methods- basic concepts-Thyristor protection circuits-snubber circuits.

Specifications of SCR- List parameters (Datasheet interpretation only)

CO2	Summarize the principle and operation of controlled rectifiers.		
M2.01	Illustrate the operation of single phase half wave controlled rectifier	4	Understanding
M2.02	Explain the operation of single phase full wave controlled rectifier	5	Understanding
M2.03	Summarize the operation of single phase semi controlled rectifiers	2	Understanding
M2.04	Outline the operation of three phase controlled rectifiers	3	Understanding
	SeriesTest1	1	
Contents: Single phase half wave controlled rectifiers – Terms and definitions – controlled rectifiers- firing angle-conduction angle-freewheeling diode -Types of controlled rectifiers (listing only) Single phase half wave controlled rectifiers-circuit diagram, waveforms and operation(R load - RL load with and without freewheeling diode) Single phase full wave controlled rectifiers -circuit diagram, waveforms and operation- bridge rectifier(with R load ,RL load only, RL load and freewheeling diode -continuous mode only) Single phase semi controlled rectifier - circuit diagram, waveforms and operation (with RL load) Three phase bridge converter with resistive load (basic diagram with concept only)			
CO3	Summarize the principle and operation of DC and AC converters.		
M3.01	Summarize the operation of DC choppers	3	Understanding
M3.02	Illustrate the classification of DC chopper	5	Understanding
M3.03	Explain the operation of buck and boost converters	2	Understanding
M3.04	Summarize the operation of Cycloconverters	2	Understanding

Contents:

DC Chopper-basic principle of operation with waveforms- equation of output voltage - simple problems- control strategies- constant frequency control - variable frequency control

Types of choppers -concept of quadrant operations- circuit diagram and basic concepts of choppers-singlequadrant(TypeA,TypeB)-twoquadrant(TypeC,TypeD)-fourquadrant (Type E)-Applications of chopper

Buck and boost converters-(operation with diagram only)

Cycloconverters –types –single phase to single phase step up cycloconverter (midpoint type)

CO4	Summarize various inverters and electric drives		
M4.01	Explain the operation of basic inverters	5	Understanding
M4.02	Illustrate the operation of bridge inverters	7	Understanding
M4.03	Explain different voltage control methods	2	Understanding
M4.04	Classify various electric drives.	2	Understanding
	Series Test II	1	

Contents:

Inverter- requirements of a practical inverter- Classification- circuit diagram and working - series inverter- parallel inverter- concept of CSI and VSI

Bridge inverters - concept of feedback diodes- single phase bridge inverter - half bridge - full bridge (circuit diagram- working wave forms for RL load). Three phase bridge inverter- 180 degree conduction mode- circuit diagram with waveforms for R loads only.

Voltage control methods - Principle of pulse width modulation- different voltage control methods-single PWM - sinusoidal PWM

Electric drives-introduction-types-advantages-basic block diagram

Text/Reference:

T/R	Book Title/Author
T1	P.S.Bimbhra, Power Electronics, Khanna Publications,Co.,New Delhi(ISBN: 9788174092151).
T2	M.D.Singhand Khanchandani K.B, Power Electronics.Secondedition, McGraw Hill Publishing Co. Ltd, New Delhi,2008,(ISBN:9780070583894)

R1	MuhammedHRashid, Power Electronics Circuits Devices and Applications, Pearson Education India, Noida, 2014. (ISBN: 9788131702468)
R2	JainAlok, Power Electronics and its Applications, Penram International Publishing, Mumbai, 2006 (ISBN: 9788187972136)
R3	Dr. BR Gupta, Singhal, Power Electronics K Kataria & Sons, 2009 (ISBN: 9788185749532)
R4	Sen P. C., Power Electronics, McGraw-Hill Publishing Company Limited, New Delhi. ISBN: 9780074624005
R5	V. R. Moorthy Oxford Publishers Power Electronics Device Circuits and Industrial Applications

Online Resources

Sl.No	Website Link
1	NPTEL >> Courses >> Electrical Engineering >> Power Electronics https://nptel.ac.in/courses/108102145 https://nptel.ac.in/courses/108101038 https://nptel.ac.in/courses/108101126 https://nptel.ac.in/courses/108108077
2	www.electrical4u.com/electrical-engineering-articles/power-electronics
3	www.swayam.gov.in
4	www.youtube.com
5	wikipedia.org
6.	IEEE papers/journals on power electronics transactions